Function: Questions and Full Solutions

Question 01

English

Let $S = \{1, 2, 3, \ldots, 10\}$ and $T = \{0, 1, 2, 3\}$. Then the number of functions $f : S \to T$ such that $f(xy) = f(x) + f(y)$ for all $x, y \in S$ with $xy \in S$ is

(A) 64
(B) 48
(C) 56
(D) 72

Only one option is correct.

हिन्दी

माना कि $S = \{1, 2, 3, \ldots, 10\}$ तथा $T = \{0, 1, 2, 3\}$ है। तब ऐसे फलनों $f : S \to T$ की संख्या, जिनके लिए $xy \in S$ वाले सभी $x, y \in S$ के लिए $f(xy) = f(x) + f(y)$ है, बराबर है

(A) 64
(B) 48
(C) 56
(D) 72

केवल एक विकल्प सही है।

Solution

Put $x = y = 1$ (allowed, since $1 \cdot 1 = 1 \in S$): $f(1) = f(1) + f(1)$, so $f(1) = 0$. Then every relation with $x = 1$ or $y = 1$ reads $f(y) = 0 + f(y)$ and gives no restriction.

The remaining products $xy \leq 10$ with $x, y \geq 2$ are $2 \cdot 2 = 4$, $2 \cdot 3 = 6$, $2 \cdot 4 = 8$, $2 \cdot 5 = 10$, $3 \cdot 3 = 9$ (and the same with the factors exchanged). Write $a = f(2)$, $b = f(3)$, $c = f(5)$.

Then $f(4) = 2a$, $f(6) = a + b$, $f(8) = f(2) + f(4) = 3a$, $f(9) = 2b$, $f(10) = a + c$. The value $f(7)$ appears in no relation, so it can be any of the 4 elements of T.

Every value must lie in $T = \{0, 1, 2, 3\}$. From $f(8) = 3a \leq 3$: $a \in \{0, 1\}$. From $f(9) = 2b \leq 3$: $b \in \{0, 1\}$; then $f(6) = a + b \leq 2$ automatically. From $f(10) = a + c \leq 3$: $c \leq 3 - a$.

Conversely, any such choice of $a, b, c, f(7)$ defines f on all of S with values in T, and all the relations hold, so the functions correspond exactly to these choices.

Case $a = 0$: b has 2 choices, c has 4, $f(7)$ has 4: 32 functions.

Case $a = 1$: b has 2 choices, $c \in \{0, 1, 2\}$ has 3, $f(7)$ has 4: 24 functions.

Total $= 32 + 24 = 56$. (Ignoring the bound on $f(10)$ gives 64; also forcing $c \leq 2$ when $a = 0$ gives 48.) A direct computer enumeration of all 4^{10} maps confirms 56.

Correct Option: (C)

Question 02

English

Let $f : (0, \infty) \to \mathbb{R}$ be a function, differentiable at $x = 1$, such that $f(xy) = f(x) + f(y)$ for all $x, y \in (0, \infty)$ and $f'(1) = 2$. Let $h : (0, \infty) \to \mathbb{R}$ be defined by $h(x) = x^2 - 3x + f(x)$. Then which of the following statements is TRUE?

(A) h is a bijection, and the equation $h(x) = h^{-1}(x)$ has exactly one solution, which lies in the interval $(3, 4)$

(B) h is a bijection, and the equation $h(x) = h^{-1}(x)$ has exactly one solution, namely $x = 1$

(C) h is a bijection, and the equation $h(x) = h^{-1}(x)$ has exactly three solutions

(D) h is not one-one

Only one option is correct.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$ एक फलन है, जो $x = 1$ पर अवकलनीय है, तथा सभी $x, y \in (0, \infty)$ के लिए $f(xy) = f(x) + f(y)$ एवं $f'(1) = 2$ है। माना कि $h : (0, \infty) \to \mathbb{R}$, $h(x) = x^2 - 3x + f(x)$ द्वारा परिभाषित है। तब निम्न में से कौन-सा कथन सत्य है?

(A) h एकैकी तथा आच्छादक है, और समीकरण $h(x) = h^{-1}(x)$ का ठीक एक हल है, जो अन्तराल $(3, 4)$ में स्थित है

(B) h एकैकी तथा आच्छादक है, और समीकरण $h(x) = h^{-1}(x)$ का ठीक एक हल है, अर्थात् $x = 1$

(C) h एकैकी तथा आच्छादक है, और समीकरण $h(x) = h^{-1}(x)$ के ठीक तीन हल हैं

(D) h एकैकी नहीं है

केवल एक विकल्प सही है।

Solution

Putting $x = y = 1$: $f(1) = 2f(1)$, so $f(1) = 0$. For $x > 0$ and small t: $f(x+t) = f\left(x\left(1 + \dfrac{t}{x}\right)\right) = f(x) + f\left(1 + \dfrac{t}{x}\right)$.

Hence $\dfrac{f(x+t) - f(x)}{t} = \dfrac{1}{x} \cdot \dfrac{f\left(1 + \frac{t}{x}\right) - f(1)}{t/x} \to \dfrac{f'(1)}{x} = \dfrac{2}{x}$ as $t \to 0$. So f is differentiable on $(0, \infty)$ with $f'(x) = \dfrac{2}{x}$, and with $f(1) = 0$ this gives $f(x) = 2\log_e x$.

Then $h(x) = x^2 - 3x + 2\log_e x$ and $h'(x) = 2x - 3 + \dfrac{2}{x} = \dfrac{2x^2 - 3x + 2}{x}$. The quadratic $2x^2 - 3x + 2$ has discriminant $9 - 16 < 0$ and positive leading coefficient, so $h'(x) > 0$ on $(0, \infty)$: h is strictly increasing and therefore one-one, so (D) is wrong.

As $x \to 0^+$, $2\log_e x \to -\infty$ while $x^2 - 3x \to 0$, so $h(x) \to -\infty$; as $x \to \infty$, $h(x) \to \infty$. Being continuous, h takes every real value: h is a bijection from $(0, \infty)$ onto $\mathbb{R}$ and $h^{-1} : \mathbb{R} \to (0, \infty)$ exists.

Let $x > 0$ satisfy $h(x) = h^{-1}(x)$ and put $y = h(x)$; then $y > 0$ and $h(y) = x$. If $x < y$, monotonicity gives $h(x) < h(y)$, i.e. $y < x$, a contradiction; if $x > y$ we get $y > x$, again a contradiction. So $y = x$. Conversely $h(x) = x$ gives $h^{-1}(x) = x = h(x)$. Thus the equation is equivalent to $h(x) = x$: the graph of an increasing function meets its mirror image in $y = x$ only on that line.

Let $g(x) = h(x) - x = x^2 - 4x + 2\log_e x$. Then $g'(x) = 2x - 4 + \dfrac{2}{x} = \dfrac{2(x-1)^2}{x} \geq 0$, with equality only at $x = 1$, so g is strictly increasing on $(0, \infty)$ and has at most one zero.

$g(3) = 9 - 12 + 2\log_e 3 = 2\log_e 3 - 3 < 0$ since $\log_e 3 < 1.5$, and $g(4) = 16 - 16 + 2\log_e 4 > 0$. By the intermediate value theorem g has exactly one zero, lying in $(3, 4)$ (numerically $x \approx 3.28$).

So $h(x) = h^{-1}(x)$ has exactly one solution, in $(3, 4)$: (A) is true. $g(1) = -3 \neq 0$, so (B) is wrong,

and three solutions (as can happen for a decreasing function) is impossible here, so (C) is wrong.

Correct Option: (A)

Question 03

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 + 2x - 2$, and let $g : \mathbb{R} \to \mathbb{R}$ be the inverse of f. Let P be a point at which the curves $y = f(x)$ and $y = g(x)$ intersect, and let θ be the acute angle between these two curves at P. Then $\tan\theta$ is equal to

(A) $\dfrac{24}{5}$

(B) $\dfrac{12}{5}$

(C) $\dfrac{5}{12}$

(D) $\dfrac{12}{13}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 + 2x - 2$ द्वारा परिभाषित है, तथा $g : \mathbb{R} \to \mathbb{R}$, f का प्रतिलोम फलन है। माना कि P एक ऐसा बिन्दु है जिस पर वक्र $y = f(x)$ तथा $y = g(x)$ प्रतिच्छेद करते हैं, तथा P पर इन दोनों वक्रों के बीच का न्यून कोण θ है। तब $\tan\theta$ बराबर है

(A) $\dfrac{24}{5}$

(B) $\dfrac{12}{5}$

(C) $\dfrac{5}{12}$

(D) $\dfrac{12}{13}$

केवल एक विकल्प सही है।

Solution

$f'(x) = 3x^2 + 2 > 0$ for all x, so f is strictly increasing and one-one; a cubic tends to $\pm\infty$ at $\pm\infty$ and is continuous, so f is onto $\mathbb{R}$. Hence $g = f^{-1} : \mathbb{R} \to \mathbb{R}$ exists.

The graph of g is the reflection of the graph of f in the line $y = x$. Suppose (a, b) lies on both curves: $b = f(a)$ and $b = g(a)$, i.e. $f(b) = a$. If $a < b$ then, f being increasing, $f(a) < f(b)$, i.e. $b < a$, a contradiction; similarly $a > b$ is impossible. So $a = b$ and P lies on $y = x$ with $f(a) = a$.

$f(x) = x$ gives $x^3 + x - 2 = 0$, i.e. $(x-1)(x^2+x+2) = 0$; the quadratic has discriminant $1 - 8 < 0$, so $x = 1$ and $P = (1, 1)$ is the only common point.

Slope of $y = f(x)$ at P: $m_1 = f'(1) = 3 + 2 = 5$. By the derivative of the inverse, $g'(1) = \dfrac{1}{f'(g(1))} = \dfrac{1}{f'(1)} = \dfrac{1}{5} = m_2$ (the reflected tangent has the reciprocal slope).

The acute angle between the tangents satisfies $\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right| = \dfrac{5 - \frac{1}{5}}{1 + 1} = \dfrac{24/5}{2} = \dfrac{12}{5}$.

Option (A) omits the denominator $1 + m_1 m_2 = 2$, (C) is $\cot\theta$, and (D) is $\sin\theta = \dfrac{12}{13}$.

Correct Option: (B)

Question 04

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \sin\left(\cos\dfrac{\pi x}{2}\right) + \cos\left(\sin\dfrac{2\pi x}{3}\right)$. Then the fundamental period of f is

(A) 6
(B) 24
(C) 4
(D) 12

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sin\left(\cos\dfrac{\pi x}{2}\right) + \cos\left(\sin\dfrac{2\pi x}{3}\right)$ द्वारा परिभाषित है। तब f का मूल आवर्तकाल है

(A) 6
(B) 24
(C) 4
(D) 12

केवल एक विकल्प सही है।

Solution

First term $u(x) = \sin\left(\cos\dfrac{\pi x}{2}\right)$. Put $t = \dfrac{\pi x}{2}$. Since $\cos(t + \pi) = -\cos t$ and sin is odd, $\sin(\cos(t + \pi)) = -\sin(\cos t)$, which is not identically equal to $\sin(\cos t)$ (at $t = 0$ the values are $-\sin 1$ and $\sin 1$). Since $\cos t$ has fundamental period 2π and sin is one-one on $[-1, 1]$, the fundamental period of $\sin(\cos t)$ is 2π; in x it is $\dfrac{2\pi}{\pi/2} = 4$.

Second term $v(x) = \cos\left(\sin\dfrac{2\pi x}{3}\right)$. Put $s = \dfrac{2\pi x}{3}$. Since $\sin(s + \pi) = -\sin s$ and cos is even, $\cos(\sin(s + \pi)) = \cos(\sin s)$. As cos is one-one on $[0, 1]$, $\cos(\sin s)$ repeats exactly when $|\sin s|$ repeats, whose fundamental period is π. In x the period is $\dfrac{\pi}{2\pi/3} = \dfrac{3}{2}$.

The periods 4 and $\dfrac{3}{2}$ are commensurable, with $\dfrac{4}{3/2} = \dfrac{8}{3}$ in lowest terms, so the period of $f = u + v$ is their LCM: $\dfrac{\text{LCM}(8, 3)}{\text{HCF}(2, 2)} = \dfrac{24}{2} = 12$; indeed $12 = 3 \cdot 4 = 8 \cdot \dfrac{3}{2}$.

No proper divisor of the form $\dfrac{12}{k}$ works. For example $f(x + 6) = -u(x) + v(x)$, which differs from $f(x)$ at $x = 0$; and $f(x + 4) = u(x) + \cos\left(\sin\left(\dfrac{2\pi x}{3} + \dfrac{8\pi}{3}\right)\right)$, which at $x = 0$ equals $\sin 1 + \cos\dfrac{\sqrt{3}}{2} \neq \sin 1 + 1$. A numerical scan of all shifts $T \in (0, 12)$ confirms that 12 is the least period.

Option (A) results from taking π as the period of $\sin(\cos t)$, and (C) is the period of the first term alone.

Correct Option: (D)

Question 05

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a function, continuous at $x = 0$, such that $f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$ and $f(x) \geq 1 + 2x$ for all $x \in \mathbb{R}$. Then the value of $f(\log_e 6)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक फलन है, जो $x = 0$ पर सतत है, तथा सभी $x, y \in \mathbb{R}$ के लिए $f(x + y) = f(x)f(y)$ एवं सभी $x \in \mathbb{R}$ के लिए $f(x) \geq 1 + 2x$ है। तब $f(\log_e 6)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

From the inequality, $f(1) \geq 3$, so f is not identically zero. If $f(a) = 0$ for some a, then $f(x) = f(a)f(x - a) = 0$ for every x, which is impossible; hence $f(x) \neq 0$ for all x.

Then $f(x) = f\left(\frac{x}{2}\right) f\left(\frac{x}{2}\right) = f\left(\frac{x}{2}\right)^2 > 0$ for all x, and $f(0) = f(0)^2$ with $f(0) \neq 0$ gives $f(0) = 1$.

Continuity everywhere: for any x, $f(x + t) = f(x)f(t)$ and $f(t) \to f(0) = 1$ as $t \to 0$ (continuity at 0), so $f(x + t) \to f(x)$; thus f is continuous on $\mathbb{R}$.

Let $g(x) = \log_e f(x)$. Then g is continuous and $g(x + y) = g(x) + g(y)$. This gives $g(n) = ng(1)$, then $g\left(\frac{p}{q}\right) = \frac{p}{q} g(1)$ for rationals, and by continuity $g(x) = kx$ for all real x, where $k = g(1)$. So $f(x) = e^{kx}$.

Now $\varphi(x) = e^{kx} - 1 - 2x \geq 0$ for all x and $\varphi(0) = 0$, so φ has a minimum at $x = 0$ and $\varphi'(0) = k - 2 = 0$, i.e. $k = 2$. (If $k > 2$, $\varphi < 0$ just left of 0; if $k < 2$, just right of 0.) Conversely $e^{2x} \geq 1 + 2x$ for all real x, since $e^t \geq 1 + t$.

Therefore $f(x) = e^{2x}$ and $f(\log_e 6) = e^{2 \log_e 6} = 6^2 = 36$.

Final Answer: 36

Question 06

English

Let $f : (0, \infty) \to \mathbb{R}$ be a function such that $f(xy) = f(x) + f(y)$ for all $x, y \in (0, \infty)$ and $\lim\limits_{x \to 1} \dfrac{f(x)}{x - 1} = 3$. Then the value of $\lim\limits_{n \to \infty} \sum\limits_{r=1}^{n} \dfrac{r}{n^2} f\left(\dfrac{r}{n}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$ एक फलन है, जिसके लिए सभी $x, y \in (0, \infty)$ के लिए $f(xy) = f(x) + f(y)$ है तथा $\lim\limits_{x \to 1} \dfrac{f(x)}{x - 1} = 3$ है। तब $\lim\limits_{n \to \infty} \sum\limits_{r=1}^{n} \dfrac{r}{n^2} f\left(\dfrac{r}{n}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Putting $x = y = 1$ gives $f(1) = 2f(1)$, so $f(1) = 0$. The given limit is then $\lim\limits_{x \to 1} \dfrac{f(x) - f(1)}{x - 1} = 3$, i.e. $f'(1) = 3$.

For $x > 0$: $f(x+t) = f\left(x\left(1 + \dfrac{t}{x}\right)\right) = f(x) + f\left(1 + \dfrac{t}{x}\right)$, so $\dfrac{f(x+t) - f(x)}{t} = \dfrac{1}{x} \cdot \dfrac{f\left(1 + \frac{t}{x}\right)}{t/x} \to \dfrac{3}{x}$ as $t \to 0$.

Thus $f'(x) = \dfrac{3}{x}$ on $(0, \infty)$ and, with $f(1) = 0$, $f(x) = 3 \log_e x$.

The sum is $\sum\limits_{r=1}^{n} \dfrac{1}{n} \cdot \dfrac{r}{n} \cdot 3 \log_e \dfrac{r}{n}$, a Riemann sum of $F(x) = 3x \log_e x$ on $[0, 1]$ with right end points. F extends continuously to $[0, 1]$ with $F(0) = 0$, since $x \log_e x \to 0$ as $x \to 0^+$; so the limit equals $\displaystyle\int_0^1 3x \log_e x \, dx$.

By parts, $\displaystyle\int_0^1 x \log_e x \, dx = \left[\frac{x^2}{2} \log_e x\right]_0^1 - \int_0^1 \frac{x^2}{2} \cdot \frac{1}{x} \, dx = 0 - \frac{1}{4} = -\frac{1}{4}$, the boundary term at 0 vanishing because $x^2 \log_e x \to 0$.

Hence the limit is $3 \cdot \left(-\dfrac{1}{4}\right) = -\dfrac{3}{4} = -0.75$ (a direct computation with $n = 200000$ gives -0.7499999).

Final Answer: -0.75

Question 07

English

Let $f : \mathbb{R} \setminus \{0, 1\} \to \mathbb{R}$ be a function such that $2f(x) + f\left(\frac{1}{1-x}\right) = 6x$ for all $x \in \mathbb{R} \setminus \{0, 1\}$. Then the value of $f(-1)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \setminus \{0, 1\} \to \mathbb{R}$ एक फलन है, जिसके लिए सभी $x \in \mathbb{R} \setminus \{0, 1\}$ के लिए $2f(x) + f\left(\frac{1}{1-x}\right) = 6x$ है। तब $f(-1)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $T(x) = \dfrac{1}{1-x}$. Then $T(T(x)) = \dfrac{1}{1-\frac{1}{1-x}} = \dfrac{x-1}{x}$ and $T(T(T(x))) = x$, so T permutes each orbit $\left\{x, \dfrac{1}{1-x}, \dfrac{x-1}{x}\right\}$ cyclically; all three points avoid 0 and 1 when $x \neq 0, 1$.

For $x = -1$: $T(-1) = \dfrac{1}{2}$, $T\left(\dfrac{1}{2}\right) = 2$, $T(2) = -1$. Write $p = f(-1)$, $q = f\left(\dfrac{1}{2}\right)$, $s = f(2)$.

The given relation at $x = -1$: $2p + q = -6$. At $x = \dfrac{1}{2}$: $2q + s = 3$. At $x = 2$: $2s + p = 12$.

From the first, $q = -6 - 2p$; then $s = 3 - 2q = 3 + 12 + 4p = 15 + 4p$; the third gives $30 + 8p + p = 12$, so $9p = -18$ and $p = -2$.

Check: $q = -2$, $s = 7$: $2(-2) + (-2) = -6$, $2(-2) + 7 = 3$, $2(7) + (-2) = 12$. (The general solution is $f(x) = \dfrac{8x}{3} - \dfrac{4}{3(1-x)} + \dfrac{2(x-1)}{3x}$, which also gives $f(-1) = -2$.)

Final Answer: -2

Question 08

English

Let D_1 be the set of all real numbers x for which $\log_e\left(\frac{2x^2-5x-7}{x^2-2x-8}\right)$ is defined, and let D_2 be the set of all real numbers x for which $\log_e\left(2x^2-5x-7\right)-\log_e\left(x^2-2x-8\right)$ is defined. Let $f: D_1 \to \mathbb{R}$ and $g: D_2 \to \mathbb{R}$ be defined by $f(x)=\log_e\left(\frac{2x^2-5x-7}{x^2-2x-8}\right)$ and $g(x)=\log_e\left(2x^2-5x-7\right)-\log_e\left(x^2-2x-8\right)$. Let S be the set of all $x \in D_1$ such that either $x \notin D_2$ or $f(x) \neq g(x)$. Then the sum of the lengths of the intervals that make up S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि D_1 उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $\log_e\left(\frac{2x^2-5x-7}{x^2-2x-8}\right)$ परिभाषित है, तथा D_2 उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $\log_e\left(2x^2-5x-7\right)-\log_e\left(x^2-2x-8\right)$ परिभाषित है। माना कि $f: D_1 \to \mathbb{R}$ तथा $g: D_2 \to \mathbb{R}$ क्रमशः $f(x)=\log_e\left(\frac{2x^2-5x-7}{x^2-2x-8}\right)$ तथा $g(x)=\log_e\left(2x^2-5x-7\right)-\log_e\left(x^2-2x-8\right)$ द्वारा परिभाषित हैं। माना कि S उन सभी $x \in D_1$ का समुच्चय है जिनके लिए या तो $x \notin D_2$ है या $f(x) \neq g(x)$ है। तब S को बनाने वाले अन्तरालों की लम्बाइयों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Factorise: $2x^2-5x-7=(2x-7)(x+1)$ and $x^2-2x-8=(x-4)(x+2)$. The critical points in increasing order are -2, -1, $\frac{7}{2}$, 4.

D_1 needs $\frac{(2x-7)(x+1)}{(x-4)(x+2)}>0$. By the sign scheme the expression is positive on $(-\infty,-2)$, negative on $(-2,-1)$, positive on $\left(-1,\frac{7}{2}\right)$, negative on $\left(\frac{7}{2},4\right)$ and positive on $(4,\infty)$. So $D_1=(-\infty,-2)\cup\left(-1,\frac{7}{2}\right)\cup(4,\infty)$.

D_2 needs both $(2x-7)(x+1)>0$, i.e. $x<-1$ or $x>\frac{7}{2}$, and $(x-4)(x+2)>0$, i.e. $x<-2$ or $x>4$. Together: $D_2=(-\infty,-2)\cup(4,\infty)$, a subset of D_1.

For $x \in D_2$, $A=2x^2-5x-7>0$ and $B=x^2-2x-8>0$, so $\log_e\frac{A}{B}=\log_e A-\log_e B$ and $f(x)=g(x)$. Hence the condition $f(x)\neq g(x)$ never holds on D_2, and f and g are identical as functions only when both are restricted to D_2.

Therefore $S=D_1\setminus D_2=\left(-1,\frac{7}{2}\right)$: there both A and B are negative, so the quotient is positive but the separate logarithms are undefined.

S is a single interval of length $\frac{7}{2}-(-1)=\frac{9}{2}=4.5$ (a Monte Carlo estimate on $[-10,10]$ gives about 4.46).

Final Answer: 4.5

Question 09

English

For a real number t, let $[t]$ denote the greatest integer less than or equal to t and let $\{t\} = t - [t]$. For a real number a, consider the relation $R_a = \left\{(x, y) \in \mathbb{R} \times \mathbb{R} : \ x = 2[y] + 3\{y\} - a\{y\}^2\right\}$. Let S be the set of all real numbers a for which R_a defines y as a function of x, that is, for which every vertical line meets R_a in at most one point. If $S = [\alpha, \beta]$, then the value of $\beta - \alpha$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

किसी वास्तविक संख्या t के लिए, $[t]$ उस महत्तम पूर्णांक को दर्शाता है जो t से छोटा या उसके बराबर है तथा $\{t\} = t - [t]$ है। किसी वास्तविक संख्या a के लिए संबंध $R_a = \left\{(x, y) \in \mathbb{R} \times \mathbb{R} : \ x = 2[y] + 3\{y\} - a\{y\}^2\right\}$ पर विचार कीजिए। माना कि S उन सभी वास्तविक संख्याओं a का समुच्चय है जिनके लिए R_a द्वारा y, x का एक फलन परिभाषित होता है, अर्थात् प्रत्येक ऊर्ध्वाधर रेखा R_a को अधिकतम एक बिन्दु पर मिलती है। यदि $S = [\alpha, \beta]$ है, तो $\beta - \alpha$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every real y is uniquely $y = n + t$ with $n = [y] \in \mathbb{Z}$ and $t = \{y\} \in [0, 1)$. Then $(x, y) \in R_a$ means $x = 2n + \varphi(t)$ with $\varphi(t) = 3t - at^2$.

R_a defines y as a function of x exactly when the map $y \mapsto 2[y] + 3\{y\} - a\{y\}^2$ is one-one on $\mathbb{R}$ (each vertical line $x = c$ then meets R_a at most once). Two things are needed: φ is one-one on $[0, 1)$ (no repetition inside a block), and the sets of values of different blocks do not overlap.

Inside a block: φ is a quadratic in t with $\varphi'(t) = 3 - 2at$ and $\varphi'(0) = 3 > 0$. If $a \le 0$, $\varphi' > 0$ on $[0, 1)$. If $a > 0$, φ increases up to $t = \dfrac{3}{2a}$ and decreases after it; it is one-one on $[0, 1)$ exactly when $\dfrac{3}{2a} \ge 1$, i.e. $a \le \dfrac{3}{2}$. For $a > \dfrac{3}{2}$ the vertex lies inside $(0, 1)$ and values just below the maximum are taken twice, so such a fail.

Between blocks: for $a \le \dfrac{3}{2}$, φ increases from $\varphi(0) = 0$ towards $\varphi(1^-) = 3 - a$, so block n gives exactly the values $x \in [2n,\ 2n + 3 - a)$. Block $n + 1$ gives $[2n + 2,\ 2n + 5 - a)$.

These are disjoint for every n if and only if $2n + 3 - a \le 2n + 2$, i.e. $a \ge 1$. If $a < 1$, the number $x = 2n + 2$ is obtained both from $y = n + 1$ and from some $y \in (n, n + 1)$, so the vertical line $x = 2n + 2$ meets R_a twice.

Hence $S = \left[1, \dfrac{3}{2}\right]$, so $\alpha = 1$, $\beta = \dfrac{3}{2}$ and $\beta - \alpha = 0.5$.

Final Answer: 0.5

Question 10

English

Let $f(x) = \sqrt{x^2 - 110x + 1000}$ and $g(x) = \sqrt{x-10}\sqrt{x-100}$, each defined on the set of all real numbers x for which its expression is a real number. Let D be the largest subset of $\mathbb{R}$ on which f and g are identical functions, and let $h : D \to \mathbb{R}$ be defined by $h(x) = x + \dfrac{2500}{x}$. Then the least value of h is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f(x) = \sqrt{x^2 - 110x + 1000}$ तथा $g(x) = \sqrt{x-10}\sqrt{x-100}$ हैं, जिनमें से प्रत्येक उन सभी वास्तविक संख्याओं x के समुच्चय पर परिभाषित है जिनके लिए उसका व्यंजक एक वास्तविक संख्या है। माना कि D, $\mathbb{R}$ का वह सबसे बड़ा उपसमुच्चय है जिस पर f तथा g सर्वसम फलन हैं, तथा $h : D \to \mathbb{R}$, $h(x) = x + \dfrac{2500}{x}$ द्वारा परिभाषित है। तब h का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $x^2 - 110x + 1000 = (x-10)(x-100)$, f is defined when $(x-10)(x-100) \geq 0$, i.e. on $(-\infty, 10] \cup [100, \infty)$.

g is defined when both $x - 10 \geq 0$ and $x - 100 \geq 0$, i.e. on $[100, \infty)$.

Two functions are identical on a set when both are defined there and agree at every point. For $x \geq 100$, both $x - 10$ and $x - 100$ are non-negative, so $\sqrt{x-10}\sqrt{x-100} = \sqrt{(x-10)(x-100)}$ and $f(x) = g(x)$. For $x \leq 10$, g is not defined. Hence $D = [100, \infty)$.

On $(0, \infty)$, AM-GM gives $x + \dfrac{2500}{x} \geq 2\sqrt{2500} = 100$, with equality only at $x = 50$. But $50 \notin D$, so this bound is not attained on D.

$h'(x) = 1 - \dfrac{2500}{x^2} = \dfrac{(x-50)(x+50)}{x^2} > 0$ for $x > 50$, so h is strictly increasing on $D = [100, \infty)$, and $h(x) \to \infty$ as $x \to \infty$.

Hence the range of h is $[h(100), \infty) = [125, \infty)$, since $h(100) = 100 + \dfrac{2500}{100} = 125$. The least value of h is 125.

Final Answer: 125

Question 11

English

For a real number t, let $[t]$ denote the greatest integer less than or equal to t. Let $f : D \to \mathbb{R}$ be defined by $f(x) = \sqrt{\dfrac{([x]+6)([x]-1)}{\log_{1/2}(|x|-3)}}$, where D is the set of all real numbers x for which this expression is a real number. Then the sum of all the integers in D is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

किसी वास्तविक संख्या t के लिए, $[t]$ उस महत्तम पूर्णांक को दर्शाता है जो t से छोटा या उसके बराबर है। माना कि $f : D \to \mathbb{R}$, $f(x) = \sqrt{\dfrac{([x]+6)([x]-1)}{\log_{1/2}(|x|-3)}}$ द्वारा परिभाषित है, जहाँ D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए यह व्यंजक एक वास्तविक संख्या है। तब D में स्थित सभी पूर्णांकों का योगफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The logarithm needs $|x| - 3 > 0$, i.e. $|x| > 3$. The fraction needs $\log_{1/2}(|x| - 3) \neq 0$, i.e. $|x| - 3 \neq 1$, so $x \neq \pm 4$.

Because the base $\dfrac{1}{2}$ is less than 1, $\log_{1/2} u > 0$ for $0 < u < 1$ and $\log_{1/2} u < 0$ for $u > 1$. So the denominator is positive for $3 < |x| < 4$ and negative for $|x| > 4$.

The square root needs the fraction to be ≥ 0. With $N = ([x]+6)([x]-1)$, this means $N \geq 0$ when $3 < |x| < 4$, and $N \leq 0$ when $|x| > 4$.

Case $3 < x < 4$: $[x] = 3$, $N = 9 \cdot 2 = 18 > 0$, so all of $(3, 4)$ is in D.

Case $-4 < x < -3$: $[x] = -4$, $N = 2 \cdot (-5) = -10 < 0$ while the denominator is positive, so this interval is excluded.

Case $|x| > 4$: $N \leq 0$ means $-6 \leq [x] \leq 1$, i.e. $-6 \leq x < 2$. Combined with $|x| > 4$ this gives $-6 \leq x < -4$ (no $x > 4$ satisfies $x < 2$). At $x = -6$, $N = 0$ and the fraction is 0, which is allowed.

So $D = [-6, -4) \cup (3, 4)$. The point $x = -4$ is excluded because there the logarithm is $\log_{1/2} 1 = 0$.

The integers in D are -6 and -5; their sum is -11.

Final Answer: -11

Question 12

English

For a real number t, let $[t]$ denote the greatest integer less than or equal to t. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \left[\dfrac{6x}{1+x^2}\right]$. Let m be the number of elements in the range of f, and let n be the number of real numbers a for which $\lim\limits_{x\to a} f(x)$ does not exist. Then the value of $m+n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

किसी वास्तविक संख्या t के लिए, $[t]$ उस महत्तम पूर्णांक को दर्शाता है जो t से छोटा या उसके बराबर है। माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \left[\dfrac{6x}{1+x^2}\right]$ द्वारा परिभाषित है। माना कि f के परिसर में अवयवों की संख्या m है, तथा उन वास्तविक संख्याओं a की संख्या n है जिनके लिए $\lim\limits_{x\to a} f(x)$ का अस्तित्व नहीं है। तब $m+n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $u(x) = \dfrac{6x}{1+x^2}$. Then $u(0) = 0$, and for $x > 0$, $u(x) = \dfrac{6}{x+\frac{1}{x}} \le \dfrac{6}{2} = 3$ by AM-GM, with equality only at $x = 1$. Since $u(-x) = -u(x)$, $-3 \le u(x) \le 3$ for all x, with $u = -3$ only at $x = -1$. As u is continuous and takes the values -3 and 3, its range is $[-3, 3]$.

Also $u'(x) = \dfrac{6(1-x^2)}{(1+x^2)^2}$, so u decreases on $(-\infty, -1]$, increases on $[-1, 1]$ and decreases on $[1, \infty)$.

Range of f: $[u(x)]$ takes every integer from -3 to 3, so the range is $\{-3, -2, -1, 0, 1, 2, 3\}$ and $m = 7$.

Since u is continuous, $f = [u]$ is constant near every point where u is not an integer, so the limit exists there. It can fail only at points where $u(x) = k \in \{-3, \ldots, 3\}$, i.e. $kx^2 - 6x + k = 0$.

$k = 0$: only $x = 0$, where u is increasing and changes sign: $\lim_{x\to 0^-} f = -1$, $\lim_{x\to 0^+} f = 0$. The limit does not exist.

$k = \pm 1, \pm 2$: the discriminant $36 - 4k^2 > 0$, so there are two distinct roots, none equal to ± 1. At each, u is strictly monotonic, so u crosses the integer k and f jumps by 1: the limit does not exist. This gives $4 \times 2 = 8$ points.

$k = 3$: only $x = 1$, a maximum of u. For x near 1, $x \ne 1$, $2 < u(x) < 3$, so $\lim_{x\to 1} f(x) = 2$ exists (it differs from $f(1) = 3$, but the limit exists).

$k = -3$: only $x = -1$, a minimum of u. For x near -1, $x \ne -1$, $-3 < u(x) < -2$, so $\lim_{x\to -1} f(x) = -3$ exists.

Hence $n = 1 + 8 = 9$ and $m + n = 7 + 9 = 16$.

Final Answer: 16

Question 13

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2 - 4x$, and let $h : \mathbb{R} \to \mathbb{R}$ be defined by $h(x) = \max\left\{f(|x|),\ \dfrac{x-5}{2}\right\}$. Then the least value of h is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 - 4x$ द्वारा परिभाषित है, तथा $h : \mathbb{R} \to \mathbb{R}$, $h(x) = \max\left\{f(|x|),\ \dfrac{x-5}{2}\right\}$ द्वारा परिभाषित है। तब h का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $x \geq 0$, $f(|x|) = x^2 - 4x$, and for $x < 0$, $f(|x|) = x^2 + 4x$. So the graph of $y = f(|x|)$ is the part of $y = x^2 - 4x$ for $x \geq 0$ together with its mirror image in the y-axis: a W-shaped curve with minimum -4 at $x = \pm 2$, passing through $(0, 0)$ and $(\pm 4, 0)$.

Let $L(x) = \dfrac{x-5}{2}$. The graph of h is the upper of the two graphs, so $h(x) \geq L(x)$ and $h(x) \geq f(|x|)$ for all x.

For $x \geq 0$: $h(x) \geq L(x) \geq L(0) = -\dfrac{5}{2}$. (At $x = 2$, $L(2) = -\dfrac{3}{2} > -4$, so the bottom of the right arc is covered by the line.)

For $x < 0$: the left arc meets the line where $x^2+4x = \dfrac{x-5}{2}$, i.e. $2x^2+7x+5 = 0$, $(2x+5)(x+1) = 0$, $x = -\dfrac{5}{2}$ or $x = -1$.

For $-\dfrac{5}{2} \leq x \leq -1$, $x^2 + 4x \leq \dfrac{x-5}{2}$ (the quadratic $2x^2 + 7x + 5 \leq 0$ there), so $h(x) = L(x)$, which is increasing; its least value on this interval is $L\left(-\dfrac{5}{2}\right) = -\dfrac{15}{4}$.

For $x < -\dfrac{5}{2}$: $h(x) \geq x^2+4x$, and x^2+4x is decreasing on $(-\infty, -2]$, so $x^2+4x > \left(-\dfrac{5}{2}\right)^2 + 4\left(-\dfrac{5}{2}\right) = -\dfrac{15}{4}$.

For $-1 < x < 0$: $h(x) \geq L(x) > L(-1) = -3$.

Combining all cases, $h(x) \geq -\dfrac{15}{4}$ with equality at $x = -\dfrac{5}{2}$. The range of h is $\left[-\dfrac{15}{4}, \infty\right)$ and the least value is -3.75. (Using $f(x)$ instead of $f(|x|)$ would give the irrational value $L\left(\dfrac{9-\sqrt{41}}{4}\right)$.)

Final Answer: -3.75

Question 14

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \lim\limits_{n\to\infty} ((4|x|)^n + |x-5|^n)^{1/n}$. Then the greatest value of the function $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \dfrac{1}{f(x)}$, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \lim\limits_{n\to\infty} ((4|x|)^n + |x-5|^n)^{1/n}$ द्वारा परिभाषित है। तब फलन $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \dfrac{1}{f(x)}$ का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $p, q \geq 0$, not both zero, and $M = \max\{p, q\} > 0$. Then $M^n \leq p^n + q^n \leq 2M^n$, so $M \leq (p^n + q^n)^{1/n} \leq 2^{1/n}M$. Since $2^{1/n} \to 1$, the limit is M.

With $p = 4|x|$ and $q = |x-5|$ (never both zero, since $x = 0$ gives $q = 5$), $f(x) = \max\{4|x|, |x-5|\}$ for every real x, and $f(x) > 0$. So $g = \dfrac{1}{f}$ is defined on $\mathbb{R}$.

For $x \leq 0$: $f(x) \geq |x-5| = 5 - x \geq 5$.

For $0 \leq x \leq 5$: $4|x| = 4x$ is increasing and $|x-5| = 5-x$ is decreasing; they are equal at $4x = 5-x$, $x = 1$. For $x < 1$, $f(x) = 5 - x > 4$; for $x \geq 1$, $f(x) \geq 4x \geq 4$. So the least value on this interval is $f(1) = 4$.

For $x \geq 5$: $f(x) \geq 4x \geq 20$.

Hence the least value of f is 4, attained at $x = 1$, and f is continuous and unbounded, so its range is $[4, \infty)$.

Then g takes exactly the values in $\left(0, \dfrac{1}{4}\right]$, and its greatest value is $\dfrac{1}{4} = 0.25$.

Final Answer: 0.25

Question 15

English

Let $f : [-3, 5] \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(t) = \begin{cases} t^2, & -3 \leq t < 0, \\ 3t + 10, & 0 \leq t \leq 5, \end{cases} \qquad g(x) = \begin{cases} x - 1, & x < 0, \\ x^2 + 2, & x \geq 0. \end{cases}$$

Let D be the set of all real numbers x for which $(f \circ g)(x)$ is defined, and let R be the set of all values of $(f \circ g)(x)$ for $x \in D$. Then the sum of all the integers in R is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [-3, 5] \to \mathbb{R}$ तथा $g : \mathbb{R} \to \mathbb{R}$ निम्न द्वारा परिभाषित हैं

$$f(t) = \begin{cases} t^2, & -3 \leq t < 0, \\ 3t + 10, & 0 \leq t \leq 5, \end{cases} \qquad g(x) = \begin{cases} x - 1, & x < 0, \\ x^2 + 2, & x \geq 0. \end{cases}$$

माना कि D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $(f \circ g)(x)$ परिभाषित है, तथा R, $x \in D$ के लिए $(f \circ g)(x)$ के सभी मानों का समुच्चय है। तब R में स्थित सभी पूर्णांकों का योगफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$(f \circ g)(x) = f(g(x))$ is defined exactly when $g(x)$ lies in the domain $[-3, 5]$ of f.

For $x < 0$: $g(x) = x - 1$, which is < -1. The condition $x - 1 \geq -3$ gives $x \geq -2$, so $x \in [-2, 0)$, and as x runs over $[-2, 0)$, $t = g(x)$ runs over $[-3, -1)$.

For $x \geq 0$: $g(x) = x^2 + 2 \geq 2$. The condition $x^2 + 2 \leq 5$ gives $0 \leq x \leq \sqrt{3}$, and $t = g(x)$ runs over $[2, 5]$.

Hence $D = [-2, 0) \cup [0, \sqrt{3}] = [-2, \sqrt{3}]$, and the values of g on D form $[-3, -1) \cup [2, 5]$. The part $[-1, 2)$ of the domain of f is never reached.

For $t \in [-3, -1)$, $f(t) = t^2$ and t^2 decreases from 9 (at $t = -3$) towards 1 (not attained), so these values form $(1, 9]$.

For $t \in [2, 5]$, $f(t) = 3t + 10$ increases from 16 to 25, giving $[16, 25]$.

So $R = (1, 9] \cup [16, 25]$. Its integers are $2, 3, \ldots, 9$, with sum $\dfrac{(2+9) \cdot 8}{2} = 44$, and $16, 17, \ldots, 25$, with sum $\dfrac{(16+25) \cdot 10}{2} = 205$.

The required sum is $44 + 205 = 249$. (Taking the whole range of f, $(0, 9] \cup [10, 25]$, would wrongly give 325.)

Final Answer: 249

Question 16

English

Let a be a real number and let $f : [-1, 0) \cup (0, 5] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} a + x^2, & -1 \leq x < 0, \\ \dfrac{\sqrt{x+4} - 2}{x}, & 0 < x \leq 5. \end{cases}$$

If $\lim\limits_{x \to 0} f(x)$ exists, then the sum of the greatest value and the least value of f is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a एक वास्तविक संख्या है तथा $f : [-1, 0) \cup (0, 5] \to \mathbb{R}$ निम्न द्वारा परिभाषित है

$$f(x) = \begin{cases} a + x^2, & -1 \leq x < 0, \\ \dfrac{\sqrt{x+4} - 2}{x}, & 0 < x \leq 5. \end{cases}$$

यदि $\lim\limits_{x \to 0} f(x)$ का अस्तित्व है, तो f के अधिकतम मान तथा न्यूनतम मान का योगफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $0 < x \leq 5$, multiplying by the conjugate, $\dfrac{\sqrt{x+4} - 2}{x} = \dfrac{(x+4) - 4}{x\left(\sqrt{x+4} + 2\right)} = \dfrac{1}{\sqrt{x+4} + 2}$. Hence $\lim\limits_{x \to 0^+} f(x) = \dfrac{1}{2+2} = \dfrac{1}{4}$.

For $-1 \le x < 0$, $\lim\limits_{x\to 0^-} f(x) = \lim\limits_{x\to 0^-} (a + x^2) = a$.

The limit at 0 exists only if the one-sided limits are equal, so $a = \dfrac{1}{4}$.

On $[-1, 0)$: $f(x) = \dfrac{1}{4} + x^2$ is decreasing (as x increases towards 0, x^2 decreases), so it takes the values $\left(\dfrac{1}{4}, \dfrac{5}{4}\right]$, with greatest value $f(-1) = \dfrac{5}{4}$; the value $\dfrac{1}{4}$ is not attained because $x = 0$ is excluded.

On $(0, 5]$: $\sqrt{x+4} + 2$ increases, so $f(x) = \dfrac{1}{\sqrt{x+4}+2}$ decreases from values close to $\dfrac{1}{4}$ down to $f(5) = \dfrac{1}{3+2} = \dfrac{1}{5}$. It takes the values $\left[\dfrac{1}{5}, \dfrac{1}{4}\right)$.

So the range of f is $\left[\dfrac{1}{5}, \dfrac{1}{4}\right) \cup \left(\dfrac{1}{4}, \dfrac{5}{4}\right]$. The greatest value is $\dfrac{5}{4}$ (at $x = -1$) and the least value is $\dfrac{1}{5}$ (at $x = 5$).

The required sum is $\dfrac{5}{4} + \dfrac{1}{5} = \dfrac{29}{20} = 1.45$.

Final Answer: 1.45

Question 17

English

Let $f : \left[-\dfrac{3}{2}, \infty\right) \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \sqrt{2x+3}$ and $g(x) = x$. Let $h(x) = \dfrac{f(x) + g(x)}{f(x) - g(x)}$, and let D denote the set of all real numbers x for which $h(x)$ is defined. Let m be the least element of D, and let p be the sum of all real numbers $x \ge m$ that do not belong to D. Then the value of $m + p$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \left[-\dfrac{3}{2}, \infty\right) \to \mathbb{R}$ तथा $g : \mathbb{R} \to \mathbb{R}$, $f(x) = \sqrt{2x+3}$ तथा $g(x) = x$ द्वारा परिभाषित हैं। माना $h(x) = \dfrac{f(x) + g(x)}{f(x) - g(x)}$ है, तथा D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $h(x)$ परिभाषित है। माना m, D का न्यूनतम अवयव है, तथा p उन सभी वास्तविक संख्याओं $x \ge m$ का योग है जो D में नहीं हैं। तब $m + p$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The quotient $h = \dfrac{f+g}{f-g}$ is defined at x exactly when x lies in the domain of f and in the domain of g, and the denominator $f(x) - g(x)$ is not zero.

The domain of f is $\left[-\dfrac{3}{2}, \infty\right)$ and the domain of g is $\mathbb{R}$, so their intersection is $\left[-\dfrac{3}{2}, \infty\right)$.

The denominator vanishes when $\sqrt{2x+3} = x$. The left side is non-negative, so a solution must have $x \ge 0$. Squaring gives $2x + 3 = x^2$, i.e. $x^2 - 2x - 3 = 0$, i.e. $(x-3)(x+1) = 0$.

Check: at $x = 3$, $\sqrt{9} = 3$, so $f(3) = g(3)$ and $3 \notin D$. At $x = -1$, $\sqrt{1} = 1 \ne -1$, so $f(-1) - g(-1) = 2 \ne 0$; the root -1 came from squaring and $-1 \in D$.

Hence $D = \left[-\dfrac{3}{2}, 3\right) \cup (3, \infty)$. Its least element is $m = -\dfrac{3}{2}$, and the only real number $x \ge m$ outside

D is 3, so $p = 3$.

Therefore $m + p = -\frac{3}{2} + 3 = \frac{3}{2} = 1.5$ (removing -1 as well would give 0.5).

Final Answer: 1.5

Question 18

English

Let $f : (-\infty, 1] \to \mathbb{R}$ and $g : \left(-\frac{5}{2}, \infty\right) \to \mathbb{R}$ be defined by $f(x) = \sqrt{4 - 2^{x+1}}$ and $g(x) = \log_e(2x+5)$. Let $h(x) = \frac{f(x)}{g(x)} + \frac{g(x)}{f(x)}$. The set of all real numbers x for which $h(x)$ is defined is of the form $(a, b) \setminus \{c\}$, where $a < c < b$. Then the value of $a + b + c$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : (-\infty, 1] \to \mathbb{R}$ तथा $g : \left(-\frac{5}{2}, \infty\right) \to \mathbb{R}$, $f(x) = \sqrt{4 - 2^{x+1}}$ तथा $g(x) = \log_e(2x + 5)$ द्वारा परिभाषित हैं। माना $h(x) = \frac{f(x)}{g(x)} + \frac{g(x)}{f(x)}$ है। उन सभी वास्तविक संख्याओं x का समुच्चय, जिनके लिए $h(x)$ परिभाषित है, $(a, b) \setminus \{c\}$ के रूप का है, जहाँ $a < c < b$ है। तब $a + b + c$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f(x) = \sqrt{4 - 2^{x+1}}$ is real when $4 - 2^{x+1} \geq 0$, i.e. $2^{x+1} \leq 2^2$. As 2^t is increasing, this means $x + 1 \leq 2$, i.e. $x \leq 1$.

$g(x) = \log_e(2x + 5)$ is defined when $2x + 5 > 0$, i.e. $x > -\frac{5}{2}$.

Both terms of h need f and g to be defined, so $x \in \left(-\frac{5}{2}, 1\right]$.

The term $\frac{f}{g}$ needs $g(x) \neq 0$. Now $\log_e(2x + 5) = 0 \iff 2x + 5 = 1 \iff x = -2$, and $-2 \in \left(-\frac{5}{2}, 1\right]$, so $x = -2$ is removed.

The term $\frac{g}{f}$ needs $f(x) \neq 0$. Now $\sqrt{4 - 2^{x+1}} = 0 \iff 2^{x+1} = 4 \iff x = 1$, so the right end point 1 is removed.

Hence h is defined exactly on $\left(-\frac{5}{2}, 1\right) \setminus \{-2\}$, which gives $a = -\frac{5}{2}$, $b = 1$, $c = -2$ (and indeed $a < c < b$).

Therefore $a + b + c = -\frac{5}{2} + 1 - 2 = -\frac{7}{2} = -3.5$.

Final Answer: -3.5

Question 19

English

The graph of $y = x^2$ is subjected to the following three transformations, in the given order: first, every point (x, y) of the graph is moved to $(2x, y)$; next, the resulting graph is reflected in the x-axis; finally, the graph so obtained is translated 3 units in the positive direction of the x-axis and 9 units in the positive direction of the y-axis. Let G be the final graph, let V be the vertex of G, and let A and B be the points where G meets the x-axis. Then the area of the triangle VAB is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्र $y = x^2$ के आलेख पर, दिए गए क्रम में, निम्न तीन रूपान्तरण किए जाते हैं: पहले, आलेख के प्रत्येक बिन्दु (x, y) को $(2x, y)$ पर ले जाया जाता है; फिर, प्राप्त आलेख का x-अक्ष में परावर्तन किया जाता है; अन्त में, इस प्रकार प्राप्त आलेख को x-अक्ष की धनात्मक दिशा में 3 इकाई तथा y-अक्ष की धनात्मक दिशा में 9 इकाई स्थानान्तरित किया जाता है। माना G अन्तिम आलेख है, V, G का शीर्ष है, तथा A और B वे बिन्दु हैं जहाँ G, x-अक्ष से मिलता है। तब त्रिभुज VAB का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

First transformation: a point (u, u^2) of $y = x^2$ goes to $(2u, u^2)$. Writing $X = 2u$, the new graph is $y = \left(\dfrac{X}{2}\right)^2 = \dfrac{X^2}{4}$, a horizontal stretch by the factor 2 (not $y = (2x)^2$).

Second transformation: reflection in the x-axis changes y to $-y$, giving $y = -\dfrac{x^2}{4}$.

Third transformation: translation by 3 units right and 9 units up replaces x by $x - 3$ and adds 9: $G:\ y = 9 - \dfrac{(x-3)^2}{4}$.

G is a downward parabola with vertex $V(3, 9)$.

On the x-axis, $9 - \dfrac{(x-3)^2}{4} = 0 \Rightarrow (x-3)^2 = 36 \Rightarrow x = -3$ or $x = 9$. So $A(-3, 0)$, $B(9, 0)$ and $AB = 12$.

The height of triangle VAB from V to AB is 9, so its area is $\dfrac{1}{2} \cdot 12 \cdot 9 = 54$.

Final Answer: 54

Question 20

English

The graph of $y = |x|$ is translated 10 units in the positive direction of the x-axis, the resulting graph is reflected in the x-axis, and the graph so obtained is translated 6 units in the positive direction of the y-axis. Let $y = f(x)$, $x \in \mathbb{R}$, be the equation of the final graph, and let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = |f(|x|)|$. Then the area of the region $\{(x, y) \in \mathbb{R}^2 : \ g(x) \le y \le 6\}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्र $y = |x|$ के आलेख को x-अक्ष की धनात्मक दिशा में 10 इकाई स्थानान्तरित किया जाता है, प्राप्त आलेख का x-अक्ष में परावर्तन किया जाता है, तथा इस प्रकार प्राप्त आलेख को y-अक्ष की धनात्मक दिशा में 6 इकाई स्थानान्तरित किया जाता है। माना अन्तिम आलेख का समीकरण $y = f(x)$, $x \in \mathbb{R}$ है, तथा $g : \mathbb{R} \to \mathbb{R}$, $g(x) = |f(|x|)|$ द्वारा परिभाषित है। तब क्षेत्र $\{(x, y) \in \mathbb{R}^2 : \ g(x) \le y \le 6\}$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Translating $y = |x|$ by 10 units right gives $y = |x - 10|$; reflecting in the x-axis gives $y = -|x - 10|$; translating 6 units up gives $f(x) = 6 - |x - 10|$.

$f(|x|)$ depends only on $|x|$, so $g(x) = |f(|x|)|$ is even and its graph is symmetric about the y-axis. It is enough to work with $x \ge 0$, where $g(x) = |f(x)|$, and double the area.

For $x \ge 0$: $f(x) = 6 - |x - 10|$ is $x - 4$ on $[0, 10]$ and $16 - x$ on $[10, \infty)$. It is negative on $[0, 4)$, zero at $x = 4$ and $x = 16$, positive in between with maximum 6 at $x = 10$, and negative for $x > 16$.

Taking the modulus: $g(x) = 4 - x$ on $[0, 4]$, $g(x) = 6 - |x - 10|$ on $[4, 16]$, and $g(x) = x - 16$ on $[16, \infty)$.

The condition $g(x) \le 6$ holds on $[0, 22]$ ($g(0) = 4$, $g \le 6$ on $[4, 16]$ with equality only at $x = 10$, and $x - 16 \le 6 \iff x \le 22$). So for $x \ge 0$ the region is $\{0 \le x \le 22, \ g(x) \le y \le 6\}$.

Its area is $\displaystyle\int_0^4 (6 - (4 - x))\, dx + \int_4^{16} (6 - (6 - |x - 10|))\, dx + \int_{16}^{22} (6 - (x - 16))\, dx$.

These are $\displaystyle\int_0^4 (2 + x)\, dx = 8 + 8 = 16$, $\displaystyle\int_4^{16} |x - 10|\, dx = 18 + 18 = 36$ and $\displaystyle\int_{16}^{22} (22 - x)\, dx = 18$, a total of 70.

By symmetry the required area is $2 \times 70 = 140$.

Final Answer: 140

Question 21

English

For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t, and let $\{t\} = t - [t]$. Let $E = \left\{x \in \mathbb{R} : \ \{x\} \ne \dfrac{1}{4}\right\}$, and let $g : E \to \mathbb{R}$ be defined by

$$g(x) = \left[\frac{1}{\{x\} - \dfrac{1}{4}}\right].$$

Let S be the set of all integers that do not belong to the range of g. Then the sum of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$t \in \mathbb{R}$ के लिए, माना $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है, तथा $\{t\} = t - [t]$ है। माना $E = \left\{x \in \mathbb{R} : \ \{x\} \neq \frac{1}{4}\right\}$ है, तथा $g : E \to \mathbb{R}$ निम्न द्वारा परिभाषित है

$$g(x) = \left[\frac{1}{\{x\} - \frac{1}{4}}\right].$$

माना S उन सभी पूर्णांकों का समुच्चय है जो g के परिसर में नहीं हैं। तब S के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $u = \{x\} - \frac{1}{4}$. Since $\{x\}$ takes every value in $[0, 1)$ and $\{x\} = \frac{1}{4}$ is excluded, u takes every value in $\left[-\frac{1}{4}, 0\right) \cup \left(0, \frac{3}{4}\right)$; the end value $u = -\frac{1}{4}$ occurs at every integer x.

The reciprocal $\frac{1}{u}$ is decreasing on each of the two pieces. On $\left[-\frac{1}{4}, 0\right)$ it runs from -4 (attained at $u = -\frac{1}{4}$) down to $-\infty$, so it takes exactly the values in $(-\infty, -4]$.

On $\left(0, \frac{3}{4}\right)$, $\frac{1}{u}$ runs from $+\infty$ down to $\frac{4}{3}$ (not attained), so it takes exactly the values in $\left(\frac{4}{3}, \infty\right)$.

Applying the greatest integer function: values in $(-\infty, -4]$ give every integer $n \leq -4$ (for instance n itself is attained, since every $n \leq -4$ lies in $(-\infty, -4]$).

Values in $\left(\frac{4}{3}, \infty\right)$ give 1 (from values in $\left(\frac{4}{3}, 2\right)$) and every integer $n \geq 2$ (from $\frac{1}{u} = n$).

No value of $\frac{1}{u}$ lies in $(-4, \frac{4}{3}]$, so $[\cdot]$ never equals $-3, -2, -1$ or 0 (these would need $\frac{1}{u} \in [-3, 1)$, which is inside the gap).

Hence the range of g is $\mathbb{Z} \setminus \{-3, -2, -1, 0\}$, $S = \{-3, -2, -1, 0\}$, and the sum of its elements is -6.

Final Answer: -6

Question 22

English

For a real number a, let $f_a : [0, \infty) \to \mathbb{R}$ be defined by $f_a(x) = 2x^2 - 3|x - a|$. Let S be the set of all real numbers a for which f_a is one-one. Then the least element of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या a के लिए, माना $f_a : [0, \infty) \to \mathbb{R}$, $f_a(x) = 2x^2 - 3|x - a|$ द्वारा परिभाषित है। माना S उन सभी वास्तविक संख्याओं a का समुच्चय है जिनके लिए f_a एकैकी है। तब S का न्यूनतम अवयव _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

f_a is continuous on the interval $[0, \infty)$. A continuous function on an interval is one-one exactly when it is strictly monotonic, so we find the intervals of monotonicity of f_a.

For $x < a$, $|x - a| = a - x$ and $f_a(x) = 2x^2 + 3x - 3a$, with $f_a'(x) = 4x + 3$. For $x \geq 0$ this is positive, so f_a is increasing on $[0, a]$ whenever $a > 0$.

For $x \geq a$, $|x-a| = x-a$ and $f_a(x) = 2x^2 - 3x + 3a$, with $f_a'(x) = 4x - 3$. This is negative for $x < \frac{3}{4}$ and positive for $x > \frac{3}{4}$.

Case $a < 0$: every $x \geq 0$ satisfies $x \geq a$, so $f_a(x) = 2x^2 - 3x + 3a$ on $[0, \infty)$. It decreases on $\left[0, \frac{3}{4}\right]$ and increases after, so, for example, $f_a(0) = f_a\left(\frac{3}{2}\right) = 3a$; f_a is not one-one.

Case $0 \leq a < \frac{3}{4}$: f_a increases on $[0, a]$, decreases on $\left[a, \frac{3}{4}\right]$ and increases on $\left[\frac{3}{4}, \infty\right)$. A continuous function that decreases and then increases takes values near $f_a\left(\frac{3}{4}\right)$ twice, so f_a is not one-one.

Case $a \geq \frac{3}{4}$: on $[0, a]$, $f_a' = 4x + 3 > 0$; on $[a, \infty)$, $f_a' = 4x - 3 \geq 4a - 3 \geq 0$ with equality at most at one point. Since f_a is continuous at $x = a$, it is strictly increasing on $[0, \infty)$ and therefore one-one. So $S = \left[\frac{3}{4}, \infty\right)$ and its least element is $\frac{3}{4} = 0.75$.

Final Answer: 0.75

Question 23

English

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4\}$. Consider the set T of all functions $f : A \to B$ such that $f(i) \neq i$ for every $i \in B$. Among the elements of T, exactly p functions have a one-one restriction to B, and exactly q functions are onto. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $A = \{1, 2, 3, 4, 5\}$ तथा $B = \{1, 2, 3, 4\}$ है। उन सभी फलनों $f : A \to B$ के समुच्चय T पर विचार कीजिए जिनके लिए प्रत्येक $i \in B$ हेतु $f(i) \neq i$ है। T के अवयवों में से ठीक p फलनों का B पर प्रतिबन्ध एकैकी है, तथा ठीक q फलन आच्छादक हैं। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Count p. The restriction $f|_B$ maps B into B; it is one-one on a finite set of the same size, so it is a bijection of B, and $f(i) \neq i$ for all $i \in B$ makes it a derangement of 4 objects: $D_4 = 4!\left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right) = 9$. The value $f(5) \in B$ is unrestricted (4 ways), so $p = 9 \cdot 4 = 36$.

Each of these 36 functions already has $f(B) = B$, so its range equals the codomain and it is onto. The remaining onto functions are those where $f|_B$ is many-one and $f(5)$ supplies the missing value.

Count q directly by inclusion-exclusion over the values of B that are not attained. Fix a set $K \subseteq B$ of k values that f avoids. Then f maps into $B \setminus K$ ($4 - k$ values).

An element $i \in B \setminus K$ must avoid itself as well, so it has $3 - k$ choices; an element $i \in K$ has all $4 - k$ choices (its forbidden value i is already excluded); the element 5 has $4 - k$ choices.

So the number of functions in T avoiding K is $N_k = (3-k)^{4-k}(4-k)^{k+1}$: $N_0 = 3^4 \cdot 4 = 324$, $N_1 = 2^3 \cdot 3^2 = 72$, $N_2 = 1 \cdot 2^3 = 8$, $N_3 = 0$, $N_4 = 0$.

$q = N_0 - \binom{4}{1}N_1 + \binom{4}{2}N_2 - \binom{4}{3}N_3 + \binom{4}{4}N_4 = 324 - 288 + 48 = 84$.

Check: $q - p = 48$ onto functions have $f|_B$ with exactly three values and $f(5)$ equal to the fourth value.

Therefore $p - q = 36 - 84 = -48$.

Final Answer: -48

Question 24

English

For a real number λ, let $f_\lambda : \mathbb{R} \to \mathbb{R}$ be defined by $f_\lambda(x) = 2x^3 + 6x^2 + \lambda x + 27$. The set of all real numbers λ for which the equation $f_\lambda(x) = 0$ has exactly one real root is an interval of the form (α, ∞). Then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या λ के लिए, माना $f_\lambda : \mathbb{R} \to \mathbb{R}$, $f_\lambda(x) = 2x^3 + 6x^2 + \lambda x + 27$ द्वारा परिभाषित है। उन सभी वास्तविक संख्याओं λ का समुच्चय, जिनके लिए समीकरण $f_\lambda(x) = 0$ का ठीक एक वास्तविक मूल है, (α, ∞) के रूप का एक अन्तराल है। तब α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f'_\lambda(x) = 6x^2 + 12x + \lambda = 6(x+1)^2 + \lambda - 6$.

If $\lambda \geq 6$, then $f'_\lambda(x) \geq 0$ with equality at most at $x = -1$, so f_λ is strictly increasing on $\mathbb{R}$ and hence one-one. As $f_\lambda(x) \to -\infty$ when $x \to -\infty$ and $f_\lambda(x) \to \infty$ when $x \to \infty$, the intermediate value theorem gives a root, and one-one-ness makes it the only one.

If $\lambda < 6$, f'_λ has two zeros $x_1 < x_2$; f_λ increases on $(-\infty, x_1]$, decreases on $[x_1, x_2]$ and increases on $[x_2, \infty)$, so it is many-one. By Rolle's theorem two roots would need a zero of f'_λ between them, so on each of the three monotonic pieces there is at most one root, and $f_\lambda = 0$ has exactly one real root exactly when the local maximum $f_\lambda(x_1)$ and the local minimum $f_\lambda(x_2)$ have the same sign.

As λ varies the number of real roots can change only when a turning value is 0, i.e. when f_λ has a double root x: $f_\lambda(x) = 0$ and $f'_\lambda(x) = 0$.

From $f'_\lambda(x) = 0$: $\lambda = -6x^2 - 12x$. Substituting, $2x^3 + 6x^2 - 6x^3 - 12x^2 + 27 = -4x^3 - 6x^2 + 27 = 0$, i.e. $4x^3 + 6x^2 - 27 = 0 = (2x-3)(2x^2 + 6x + 9)$. The quadratic factor has discriminant $36 - 72 < 0$, so $x = \dfrac{3}{2}$ and $\lambda = -6 \cdot \dfrac{9}{4} - 18 = -\dfrac{63}{2}$.

So the root count is constant on $\left(-\dfrac{63}{2}, \infty\right)$ and on $\left(-\infty, -\dfrac{63}{2}\right)$. At $\lambda = 0$ the turning points are $x = 0$ and $x = -2$ with $f_0(0) = 27 > 0$ and $f_0(-2) = 35 > 0$, so there is exactly one root; hence exactly one root for every $\lambda > -\dfrac{63}{2}$.

At $\lambda = -\dfrac{63}{2}$, $f_\lambda(x) = 2\left(x - \dfrac{3}{2}\right)^2 (x+6)$ has two distinct real roots, and for $\lambda < -\dfrac{63}{2}$ (e.g. $\lambda = -40$: $f(-2) > 0$, $f(2) = -5 < 0$) there are three.

Therefore the required set is $\left(-\dfrac{63}{2}, \infty\right)$ and $\alpha = -31.5$ (the one-one condition alone would wrongly give 6).

Final Answer: -31.5

Question 25

English

Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 2, 3, 4\}$. Then the number of functions $f : A \to B$ such that $f(1) < f(2)$ and the range of f is a proper subset of B is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A = \{1, 2, 3, 4, 5, 6\}$ तथा $B = \{1, 2, 3, 4\}$ है। तब उन फलनों $f : A \to B$ की संख्या, जिनके लिए $f(1) < f(2)$ है तथा f का परिसर B का एक उचित उपसमुच्चय है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The range of f is a proper subset of B exactly when f is not onto. So N is the number of functions with $f(1) < f(2)$ minus the number of onto functions with $f(1) < f(2)$.

Functions with $f(1) < f(2)$: the pair $\{f(1), f(2)\}$ is a 2-element subset of B placed in increasing order, in $\binom{4}{2} = 6$ ways, and $f(3), f(4), f(5), f(6)$ are free, in $4^4 = 256$ ways. Total $6 \cdot 256 = 1536$.

Onto functions $A \to B$ (inclusion-exclusion over the missed values): $4^6 - \binom{4}{1}3^6 + \binom{4}{2}2^6 - \binom{4}{3}1^6 = 4096 - 2916 + 384 - 4 = 1560$.

Onto functions with $f(1) = f(2)$: treat 1 and 2 as one element, so these are onto functions from a 5-element set to B: $4^5 - 4 \cdot 3^5 + 6 \cdot 2^5 - 4 = 1024 - 972 + 192 - 4 = 240$.

For an onto f with $f(1) \neq f(2)$, interchanging the values $f(1)$ and $f(2)$ gives another onto function with the opposite inequality. Hence the onto functions with $f(1) \neq f(2)$ split equally: onto with $f(1) < f(2)$ is $\dfrac{1560 - 240}{2} = 660$.

Therefore $N = 1536 - 660 = 876$.

Check by direct inclusion-exclusion on not-onto functions with $f(1) < f(2)$: missing one given value leaves $\binom{3}{2}3^4 = 243$ functions, missing two given values leaves $\binom{2}{2}2^4 = 16$, missing three leaves none; so $N = 4 \cdot 243 - 6 \cdot 16 = 972 - 96 = 876$.

Final Answer: 876

Question 26

English

Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} -\frac{x}{8}, & x < 0 \\ x^2 - x + \frac{1}{2}, & 0 \le x \le 1 \\ \frac{3}{2} - x, & x > 1 \end{cases} \qquad g(x) = \begin{cases} x + 1, & x < \frac{1}{4} \\ 2x, & x \ge \frac{1}{4} \end{cases}$$

Let $h : \mathbb{R} \to \mathbb{R}$ be defined by $h(x) = g(f(x))$, and let S be the set of all points of $\mathbb{R}$ at which h is NOT differentiable. Then the sum of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ तथा $g : \mathbb{R} \to \mathbb{R}$ इस प्रकार परिभाषित हैं

$$f(x) = \begin{cases} -\frac{x}{8}, & x < 0 \\ x^2 - x + \frac{1}{2}, & 0 \le x \le 1 \\ \frac{3}{2} - x, & x > 1 \end{cases} \qquad g(x) = \begin{cases} x + 1, & x < \frac{1}{4} \\ 2x, & x \ge \frac{1}{4} \end{cases}$$

माना $h : \mathbb{R} \to \mathbb{R}$, $h(x) = g(f(x))$ द्वारा परिभाषित है, तथा S, $\mathbb{R}$ के उन सभी बिन्दुओं का समुच्चय है जिन पर h अवकलनीय नहीं है। तब S के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The rule for $g(t)$ changes only at $t = \frac{1}{4}$, and g is discontinuous there ($g(\frac{1}{4}^-) = \frac{5}{4}$, $g(\frac{1}{4}) = \frac{1}{2}$). So $h = g \circ f$ can fail to be smooth only where f changes its rule ($x = 0$, $x = 1$) or where $f(x)$ crosses $\frac{1}{4}$.

For $x < 0$: $f(x) = -\frac{x}{8} > 0$ and $f(x) < \frac{1}{4} \iff x > -2$. Hence $h(x) = 2f(x) = -\frac{x}{4}$ for $x \le -2$ and $h(x) = f(x) + 1 = 1 - \frac{x}{8}$ for $-2 < x < 0$.

For $0 \le x \le 1$: $f(x) = x^2 - x + \frac{1}{2} = \left(x - \frac{1}{2}\right)^2 + \frac{1}{4} \ge \frac{1}{4}$, with equality only at $x = \frac{1}{2}$. So $h(x) = 2f(x) = 2x^2 - 2x + 1$ on the whole interval. At $x = \frac{1}{2}$ the value of f touches $\frac{1}{4}$ from above and never goes below it, so h uses the same rule on both sides and is differentiable there.

For $x > 1$: $f(x) = \frac{3}{2} - x$, and $f(x) \ge \frac{1}{4} \iff x \le \frac{5}{4}$. So $h(x) = 2f(x) = 3 - 2x$ on $\left(1, \frac{5}{4}\right]$ and $h(x) = f(x) + 1 = \frac{5}{2} - x$ for $x > \frac{5}{4}$.

At $x = -2$: $h(-2) = \frac{1}{2}$ and $\lim_{x \to -2^-} h(x) = \frac{1}{2}$, but $\lim_{x \to -2^+} h(x) = 1 + \frac{1}{4} = \frac{5}{4}$. So h is discontinuous, hence not differentiable, at -2.

At $x = 0$: f itself is discontinuous ($f(0^-) = 0$, $f(0) = \frac{1}{2}$), yet $\lim_{x \to 0^-} h(x) = 1$ and $h(0) = 2 \cdot \frac{1}{2} = 1 = \lim_{x \to 0^+} h(x)$, so h is continuous at 0. The left derivative is $-\frac{1}{8}$ and the right derivative is $4x - 2\big|_{x=0} = -2$, so h is not differentiable at 0.

At $x = 1$: $h(1) = 1$ and $\lim_{x\to 1^+}(3-2x) = 1$, so h is continuous; the left derivative is $4x-2\big|_{x=1} = 2$ and the right derivative is -2, so h is not differentiable at 1.

At $x = \dfrac{5}{4}$: $h\left(\dfrac{5}{4}\right) = \dfrac{1}{2}$ and the left limit is $\dfrac{1}{2}$, but the right limit is $\dfrac{5}{2} - \dfrac{5}{4} = \dfrac{5}{4}$. So h is discontinuous at $\dfrac{5}{4}$.

Every piece of h is a polynomial, so h is differentiable at all other points, including $x = \dfrac{1}{2}$.

Hence $S = \left\{-2, 0, 1, \dfrac{5}{4}\right\}$ and the sum of its elements is $-2 + 0 + 1 + \dfrac{5}{4} = \dfrac{1}{4} = 0.25$.

Final Answer: 0.25

Question 27

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \dfrac{5}{4} - 4\left(x - \dfrac{5}{4}\right)^3$. If f^{-1} is the inverse function of f, then the sum of all real numbers x satisfying $f(x) = f^{-1}(x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{5}{4} - 4\left(x - \dfrac{5}{4}\right)^3$ द्वारा परिभाषित है। यदि f^{-1}, f का प्रतिलोम फलन है, तो समीकरण $f(x) = f^{-1}(x)$ को सन्तुष्ट करने वाली सभी वास्तविक संख्याओं x का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f'(x) = -12\left(x - \dfrac{5}{4}\right)^2 \le 0$, with equality only at $x = \dfrac{5}{4}$, so f is strictly decreasing, hence one-one.

It is a cubic with negative leading coefficient, so $f(x) \to \mp\infty$ as $x \to \pm\infty$ and, being continuous, f is onto $\mathbb{R}$. Thus $f^{-1} : \mathbb{R} \to \mathbb{R}$ exists, and its graph is the reflection of the graph of f in $y = x$.

For an increasing function the two graphs can meet only on $y = x$; here f is decreasing, so common points off $y = x$ are possible and must be looked for.

If $f(x) = f^{-1}(x) = y$, then $y = f(x)$ and $f(y) = x$, so $f(f(x)) = x$. Conversely, if $f(f(x)) = x$, apply f^{-1}: $f(x) = f^{-1}(x)$. So the required x are exactly the solutions of $f(f(x)) = x$.

Let $u = x - \dfrac{5}{4}$. Then $f(x) - \dfrac{5}{4} = -4u^3$, so in the variable u the map f is $u \mapsto -4u^3$. Applying it twice: $u \mapsto -4(-4u^3)^3 = 256u^9$.

So $f(f(x)) = x \iff 256u^9 = u \iff u\left(256u^8 - 1\right) = 0 \iff u = 0$ or $u^8 = \dfrac{1}{256} = \left(\dfrac{1}{2}\right)^8$. The real solutions are $u = 0,\ \dfrac{1}{2},\ -\dfrac{1}{2}$.

Hence $x = \dfrac{5}{4},\ \dfrac{7}{4},\ \dfrac{3}{4}$. Check: $f\left(\dfrac{5}{4}\right) = \dfrac{5}{4}$; $f\left(\dfrac{7}{4}\right) = \dfrac{5}{4} - 4 \cdot \dfrac{1}{8} = \dfrac{3}{4}$ and $f\left(\dfrac{3}{4}\right) = \dfrac{5}{4} + \dfrac{1}{2} = \dfrac{7}{4}$, so $f^{-1}\left(\dfrac{7}{4}\right) = \dfrac{3}{4} = f\left(\dfrac{7}{4}\right)$ and $f^{-1}\left(\dfrac{3}{4}\right) = \dfrac{7}{4} = f\left(\dfrac{3}{4}\right)$.

The sum is $\dfrac{5}{4} + \dfrac{7}{4} + \dfrac{3}{4} = \dfrac{15}{4} = 3.75$. Solving only $f(x) = x$ would give the single root $\dfrac{5}{4}$ and miss the symmetric pair $\left(\dfrac{3}{4}, \dfrac{7}{4}\right), \left(\dfrac{7}{4}, \dfrac{3}{4}\right)$.

Final Answer: 3.75

Question 28

English

Let $f : \mathbb{R} \to \mathbb{R}$ be an invertible polynomial function with inverse function f^{-1}, such that $f(x) + 2f^{-1}(x) = 3x$ for all $x \in \mathbb{R}$ and $f(1) = 7$. Then the value of $f(5) + f^{-1}(41)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक व्युत्क्रमणीय बहुपद फलन है जिसका प्रतिलोम फलन f^{-1} है, तथा सभी $x \in \mathbb{R}$ के लिए $f(x) + 2f^{-1}(x) = 3x$ है और $f(1) = 7$ है। तब $f(5) + f^{-1}(41)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

f is invertible, so it is onto $\mathbb{R}$ and $f^{-1}(f(t)) = t$ for every t. Putting $x = f(t)$ in the given relation: $f(f(t)) + 2t = 3f(t)$ for all $t \in \mathbb{R}$.

Let $d = \deg f \geq 1$ (a constant is not invertible). The left side has degree d^2 (the term $2t$ cannot cancel a degree $d^2 \geq 1$ term unless $d = 1$), and the right side has degree d. So $d^2 = d$, i.e. $d = 1$, and $f(t) = at + b$ with $a \neq 0$.

Then $f(f(t)) = a(at+b)+b = a^2t+ab+b$, and the relation becomes $a^2t+ab+b+2t = 3at+3b$ for all t. Comparing coefficients: $a^2 - 3a + 2 = 0$, so $a = 1$ or $a = 2$; and $ab + b = 3b$, i.e. $b(a - 2) = 0$.

If $a = 1$, then $b = 0$ and $f(x) = x$, which gives $f(1) = 1 \neq 7$. So $a = 2$, b is free from the relation, and $f(1) = 2 + b = 7$ gives $b = 5$.

Thus $f(x) = 2x + 5$ and $f^{-1}(x) = \dfrac{x-5}{2}$ (its graph is the mirror image of $y = 2x + 5$ in $y = x$).

Check: $f(x) + 2f^{-1}(x) = 2x + 5 + x - 5 = 3x$.

$f(5) = 15$ and $f^{-1}(41) = \dfrac{36}{2} = 18$, so $f(5) + f^{-1}(41) = 33$.

Final Answer: 33

Question 29

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \dfrac{x^3 + x}{5}$. If f^{-1} is the inverse function of f, then the total area of the regions enclosed between the curves $y = f(x)$ and $y = f^{-1}(x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x^3 + x}{5}$ द्वारा परिभाषित है। यदि f^{-1}, f का प्रतिलोम फलन है, तो वक्रों $y = f(x)$ तथा $y = f^{-1}(x)$ के बीच घिरे क्षेत्रों का कुल क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f'(x) = \dfrac{3x^2 + 1}{5} > 0$ for all x, so f is strictly increasing; as a cubic with positive leading coefficient it is onto $\mathbb{R}$. Hence $f^{-1} : \mathbb{R} \to \mathbb{R}$ exists, and the graph of $y = f^{-1}(x)$ is the reflection of the graph of $y = f(x)$ in the line $y = x$.

For a strictly increasing f, if $f(a) = f^{-1}(a) = b$ then $f(b) = a$; $a < b$ would give $f(a) < f(b)$, i.e. $b < a$, and similarly $a > b$ is impossible. So the curves meet only on $y = x$, where $\dfrac{x^3 + x}{5} = x$, i.e. $x^3 - 4x = 0$, $x = -2, 0, 2$.

On $(0, 2)$: $x - f(x) = \dfrac{4x - x^3}{5} = \dfrac{x(2 - x)(2 + x)}{5} > 0$, so $y = f(x)$ lies below $y = x$ and, by reflection, $y = f^{-1}(x)$ lies above it. The loop between the curves over $[0, 2]$ is symmetric about $y = x$, so its area is twice the area between $y = x$ and $y = f(x)$.

$\displaystyle\int_0^2 \left(x - \frac{x^3 + x}{5}\right) dx = \frac{1}{5}\int_0^2 \left(4x - x^3\right) dx = \frac{1}{5}\left(8 - 4\right) = \frac{4}{5}$, so this loop has area $\dfrac{8}{5}$. (No formula for f^{-1} is needed.)

f is odd, so f^{-1} is odd as well, and the whole picture is symmetric about the origin: the loop over $[-2, 0]$ also has area $\dfrac{8}{5}$.

Total area $= \dfrac{8}{5} + \dfrac{8}{5} = \dfrac{16}{5} = 3.2$. (Numerical integration of $\left|f(x) - f^{-1}(x)\right|$ over $[-2, 2]$ also gives 3.2.)

Final Answer: 3.2

Question 30

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 9\left\{\frac{x}{3}\right\}^2 - 6\left\{\frac{x}{3}\right\}$, where $\{t\} = t - [t]$ and $[t]$ denotes the greatest integer less than or equal to t. Let $g : \mathbb{R} \to \mathbb{R}$ be an even function and $h : \mathbb{R} \to \mathbb{R}$ be an odd function such that $f(x) = g(x) + h(x)$ for all $x \in \mathbb{R}$. Then the value of $h\left(\frac{37}{4}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 9\left\{\frac{x}{3}\right\}^2 - 6\left\{\frac{x}{3}\right\}$ द्वारा परिभाषित है, जहाँ $\{t\} = t - [t]$ तथा $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। माना $g : \mathbb{R} \to \mathbb{R}$ एक सम फलन तथा $h : \mathbb{R} \to \mathbb{R}$ एक विषम फलन है जिनके लिए सभी $x \in \mathbb{R}$ के लिए $f(x) = g(x) + h(x)$ है। तब $h\left(\frac{37}{4}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If $f = g + h$ with g even and h odd, then $f(-x) = g(x) - h(x)$, so $g(x) = \dfrac{f(x) + f(-x)}{2}$ and $h(x) = \dfrac{f(x) - f(-x)}{2}$; the decomposition is unique.

Let $s = 3\left\{\frac{x}{3}\right\}$. Then $0 \le s < 3$, $x - s$ is a multiple of 3, and $f(x) = s^2 - 2s$. Since $\left\{\dfrac{x+3}{3}\right\} = \left\{\dfrac{x}{3}\right\}$, $f(x+3) = f(x)$. No smaller positive period exists: a period $p \in (0,3)$ would need $f(p) = f(0) = 0$, so $p = 2$, but $f(1) = -1 \neq 0 = f(3)$. So 3 is the fundamental period and $f(x) = f(x - 3k)$ for every integer k.

$\dfrac{37}{4} - 9 = \dfrac{1}{4} \in [0,3)$, so $f\left(\dfrac{37}{4}\right) = f\left(\dfrac{1}{4}\right) = \dfrac{1}{16} - \dfrac{1}{2} = -\dfrac{7}{16}$.

$-\dfrac{37}{4} + 12 = \dfrac{11}{4} \in [0,3)$, so $f\left(-\dfrac{37}{4}\right) = f\left(\dfrac{11}{4}\right) = \dfrac{121}{16} - \dfrac{11}{2} = \dfrac{33}{16}$.

Therefore $h\left(\dfrac{37}{4}\right) = \dfrac{1}{2}\left(-\dfrac{7}{16} - \dfrac{33}{16}\right) = \dfrac{1}{2}\cdot\left(-\dfrac{40}{16}\right) = -\dfrac{5}{4} = -1.25$. (For comparison, $g\left(\dfrac{37}{4}\right) = \dfrac{13}{16}$; the trap $f(-x) = -f(x)$ would give $-\dfrac{7}{16}$.)

Final Answer: -1.25

Question 31

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that the function $g : \mathbb{R} \to \mathbb{R}$ defined by $g(x) = (x+1)f(x)$ is an odd function, and

$$\int_{-x}^{x} f(t)\,dt = -\frac{2x^5}{5} \quad \text{for all } x \in \mathbb{R}.$$

Then the value of $f(3)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक सतत फलन है, इस प्रकार कि $g(x) = (x+1)f(x)$ द्वारा परिभाषित फलन $g : \mathbb{R} \to \mathbb{R}$ एक

विषम फलन है, तथा सभी $x \in \mathbb{R}$ के लिए

$$\int_{-x}^{x} f(t)\,dt = -\frac{2x^5}{5}$$

है। तब $f(3)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every $f : \mathbb{R} \to \mathbb{R}$ is uniquely $f = E + O$ with $E(x) = \dfrac{f(x) + f(-x)}{2}$ even and $O(x) = \dfrac{f(x) - f(-x)}{2}$ odd; both are continuous because f is.

$g(x) = (x+1)f(x) = xE(x) + xO(x) + E(x) + O(x)$. The product of the odd function x with an even function is odd and with an odd function is even. So the odd part of g is $xE(x) + O(x)$ and the even part of g is $xO(x) + E(x)$.

g is odd, so its even part vanishes: $E(x) = -xO(x)$ for all x.

For an odd continuous function $\displaystyle\int_{-x}^{x} O(t)\,dt = 0$, and for an even one $\displaystyle\int_{-x}^{x} E(t)\,dt = 2\int_{0}^{x} E(t)\,dt$.

Hence $\displaystyle 2\int_{0}^{x} E(t)\,dt = -\frac{2x^5}{5}$ for all x; differentiating, $2E(x) = -2x^4$, so $E(x) = -x^4$.

Then $-xO(x) = -x^4$ gives $O(x) = x^3$ for $x \neq 0$, and $O(0) = 0$ for any odd function, so $O(x) = x^3$ for all x.

Therefore $f(x) = x^3 - x^4$. Check: $(x+1)(x^3 - x^4) = x^3 - x^5$ is odd, and $\displaystyle\int_{-x}^{x} (t^3 - t^4)\,dt = 0 - \frac{2x^5}{5}$.

$f(3) = 27 - 81 = -54$.

Final Answer: -54

Question 32

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2\cos^2\left(\dfrac{\pi x}{6}\right) + \sin\left(\dfrac{\pi x}{4}\right)$. Let $g : \mathbb{R} \to \mathbb{R}$ be an even function and $h : \mathbb{R} \to \mathbb{R}$ be an odd function such that $f(x) = g(x) + h(x)$ for all $x \in \mathbb{R}$. Let T_1, T_2 and T_3 be the fundamental periods of g, h and ϕ, respectively, where $\phi : \mathbb{R} \to \mathbb{R}$ is defined by $\phi(x) = f(x)\,f(-x)$. Then the value of $T_1 + T_2 + T_3$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2\cos^2\left(\dfrac{\pi x}{6}\right) + \sin\left(\dfrac{\pi x}{4}\right)$ द्वारा परिभाषित है। माना $g : \mathbb{R} \to \mathbb{R}$ एक सम फलन तथा $h : \mathbb{R} \to \mathbb{R}$ एक विषम फलन इस प्रकार हैं कि सभी $x \in \mathbb{R}$ के लिए $f(x) = g(x) + h(x)$ है। माना T_1, T_2 तथा T_3 क्रमशः g, h तथा ϕ के मूल आवर्तकाल हैं, जहाँ $\phi : \mathbb{R} \to \mathbb{R}$, $\phi(x) = f(x)\,f(-x)$ द्वारा परिभाषित है। तब $T_1 + T_2 + T_3$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The decomposition of f into an even and an odd part is unique: $g(x) = \dfrac{f(x) + f(-x)}{2}$ and $h(x) = \dfrac{f(x) - f(-x)}{2}$.

Since $\cos^2$ is even and sin is odd, $f(-x) = 2\cos^2\dfrac{\pi x}{6} - \sin\dfrac{\pi x}{4}$. Hence $g(x) = 2\cos^2\dfrac{\pi x}{6} = 1 + \cos\dfrac{\pi x}{3}$

and $h(x) = \sin\dfrac{\pi x}{4}$.

The period of $\cos\dfrac{\pi x}{3}$ is $\dfrac{2\pi}{\pi/3} = 6$, so $T_1 = 6$. Note that $\cos\dfrac{\pi x}{6}$ has period 12, but its square has period 6, because $\cos^2 t$ has period π.

The period of $\sin\dfrac{\pi x}{4}$ is $\dfrac{2\pi}{\pi/4} = 8$, so $T_2 = 8$.

Since $f = g + h$ and $f(-x) = g(x) - h(x)$, $\phi(x) = (g+h)(g-h) = g^2 - h^2$.

$g^2 = 1 + 2\cos\dfrac{\pi x}{3} + \cos^2\dfrac{\pi x}{3} = \dfrac{3}{2} + 2\cos\dfrac{\pi x}{3} + \dfrac{1}{2}\cos\dfrac{2\pi x}{3}$ and $h^2 = \dfrac{1}{2} - \dfrac{1}{2}\cos\dfrac{\pi x}{2}$.

Therefore $\phi(x) = 1 + 2\cos\dfrac{\pi x}{3} + \dfrac{1}{2}\cos\dfrac{2\pi x}{3} + \dfrac{1}{2}\cos\dfrac{\pi x}{2}$. The three cosine terms have periods 6, 3 and 4.

Cosines with different frequencies cannot cancel one another, so $T > 0$ is a period of ϕ only if it is a period of each term, that is, a common multiple of 6, 3 and 4. The least such number is 12, and $\phi(x + 12) = \phi(x)$, so $T_3 = 12$. (Taking the period of f, which is $\mathrm{LCM}(6, 8) = 24$, would be wrong: in ϕ only h^2 appears, whose period is 4.)

Hence $T_1 + T_2 + T_3 = 6 + 8 + 12 = 26$.

Final Answer: 26

Question 33

English

Let $f : \mathbb{R} \to \mathbb{R} \setminus \{-1, 1\}$ be a function such that $f(x+1) = \dfrac{f(x) - 3}{f(x) + 1}$ for all $x \in \mathbb{R}$, and $f(11) = -\dfrac{9}{7}$. Then the value of $f(2026)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $f : \mathbb{R} \to \mathbb{R} \setminus \{-1, 1\}$ एक ऐसा फलन है कि सभी $x \in \mathbb{R}$ के लिए $f(x+1) = \dfrac{f(x) - 3}{f(x) + 1}$ है, तथा $f(11) = -\dfrac{9}{7}$ है। तब $f(2026)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $\varphi(y) = \dfrac{y-3}{y+1}$, so that $f(x+1) = \varphi(f(x))$. The value $\varphi(y)$ is never 1, and it is -1 only for $y = 1$, so φ maps $\mathbb{R} \setminus \{-1, 1\}$ into itself and the relation can be iterated.

$f(x+2) = \varphi(\varphi(f(x)))$, where $\varphi(\varphi(y)) = \dfrac{\frac{y-3}{y+1} - 3}{\frac{y-3}{y+1} + 1} = \dfrac{(y-3) - 3(y+1)}{(y-3) + (y+1)} = \dfrac{-2y-6}{2y-2} = -\dfrac{y+3}{y-1}$.

$f(x+3) = \varphi(\varphi(\varphi(f(x))))$, where $\varphi\left(-\dfrac{y+3}{y-1}\right) = \dfrac{-(y+3) - 3(y-1)}{-(y+3) + (y-1)} = \dfrac{-4y}{-4} = y$.

Hence $f(x+3) = f(x)$ for all $x \in \mathbb{R}$: f is periodic with period 3. The periods 1 and 2 are impossible, because $\varphi(y) = y$ gives $y^2 + y = y - 3$, i.e. $y^2 = -3$, and $\varphi(\varphi(y)) = y$ gives $-(y+3) = y^2 - y$, i.e. $y^2 = -3$; neither has a real root. So 3 is the least positive integer period.

For a function with period 3, $f(x + 3k) = f(x)$ for every integer k. Since $2026 = 3 \cdot 675 + 1$, $f(2026) = f(1)$; since $11 = 3 \cdot 3 + 2$, $f(2) = f(11) = -\dfrac{9}{7}$.

Now $f(2) = \varphi(f(1))$, so $f(1) = \varphi^{-1}(f(2))$. As φ^3 is the identity, $\varphi^{-1} = \varphi \circ \varphi$, and $f(1) =$

$$-\frac{f(2)+3}{f(2)-1} = -\frac{-\frac{9}{7}+3}{-\frac{9}{7}-1} = -\frac{12/7}{-16/7} = \frac{3}{4}.$$

Check: $\varphi\left(\frac{3}{4}\right) = \frac{\frac{3}{4}-3}{\frac{3}{4}+1} = \frac{-9/4}{7/4} = -\frac{9}{7} = f(2)$.

Therefore $f(2026) = \frac{3}{4} = 0.75$.

Final Answer: 0.75

Question 34

English

For $x \geq 0$, let θ_x denote the radian measure of the angle subtended at the centre of a circle of radius $\frac{40}{\pi}$ cm by an arc of length x cm of the circle. Let $f : [0, \infty) \to \mathbb{R}$ be defined by $f(x) = \sin^2(x^\circ) + |\cos\theta_x|$, where x° denotes an angle of x degrees. Then the fundamental period of f is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$x \geq 0$ के लिए, माना θ_x उस कोण की रेडियन माप है जो $\frac{40}{\pi}$ cm त्रिज्या वाले वृत्त के x cm लम्बाई के चाप द्वारा वृत्त के केन्द्र पर अंतरित होता है। माना $f : [0, \infty) \to \mathbb{R}$, $f(x) = \sin^2(x^\circ) + |\cos\theta_x|$ द्वारा परिभाषित है, जहाँ x° का अर्थ x अंश का कोण है। तब f का मूल आवर्तकाल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Degree to radian: $x^\circ = \frac{\pi x}{180}$ radian. Hence $\sin^2(x^\circ) = \frac{1-\cos(2x^\circ)}{2} = \frac{1}{2} - \frac{1}{2}\cos\frac{\pi x}{90}$, whose fundamental period is $\frac{2\pi}{\pi/90} = 180$ (half of the period 360 of $\sin(x^\circ)$).

An arc of length l on a circle of radius r subtends $\theta = \frac{l}{r}$ radian at the centre. So $\theta_x = \frac{x}{40/\pi} = \frac{\pi x}{40}$.

$\cos\frac{\pi x}{40}$ has period 80, but $|\cos t|$ has fundamental period π, so $\left|\cos\frac{\pi x}{40}\right|$ has fundamental period $\frac{\pi}{\pi/40} = 40$.

Write $f = u + v$ with $u(x) = \sin^2(x^\circ)$ and $v(x) = \left|\cos\frac{\pi x}{40}\right|$. If $f(x+T) = f(x)$ for all $x \geq 0$, then $u(x+T) - u(x) = v(x) - v(x+T)$. The left side is a combination of $\cos\frac{\pi x}{90}$ and $\sin\frac{\pi x}{90}$; the right side, like v, repeats every 40. A non-zero combination of $\cos\frac{\pi x}{90}$, $\sin\frac{\pi x}{90}$ does not repeat every 40 (that would need $\frac{40}{180}$ to be an integer), so both sides are 0: T is a period of u and of v.

Hence T is a common multiple of 180 and 40, and the least one is $\text{LCM}(180, 40) = 360$. Indeed $f(x+360) = f(x)$, as $360 = 2 \cdot 180 = 9 \cdot 40$.

Therefore the fundamental period of f is 360.

Final Answer: 360

Question 35

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \left(\sin\left(\frac{\pi x}{10} + \frac{\pi}{6}\right) + \sin\left(\frac{\pi x}{10} - \frac{\pi}{6}\right)\right)\left(\cos\left(\frac{\pi x}{30} - \frac{\pi}{3}\right) - \cos\left(\frac{\pi x}{30} + \frac{\pi}{3}\right)\right).$$

Then the fundamental period of f is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $f : \mathbb{R} \to \mathbb{R}$

$$f(x) = \left(\sin\left(\frac{\pi x}{10} + \frac{\pi}{6}\right) + \sin\left(\frac{\pi x}{10} - \frac{\pi}{6}\right)\right)\left(\cos\left(\frac{\pi x}{30} - \frac{\pi}{3}\right) - \cos\left(\frac{\pi x}{30} + \frac{\pi}{3}\right)\right)$$

द्वारा परिभाषित है। तब f का मूल आवर्तकाल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the compound-angle formulas, $\sin(A+B) + \sin(A-B) = 2\sin A\cos B$. With $A = \dfrac{\pi x}{10}$, $B = \dfrac{\pi}{6}$ the first factor is $2 \cdot \dfrac{\sqrt{3}}{2}\sin\dfrac{\pi x}{10} = \sqrt{3}\sin\dfrac{\pi x}{10}$, of period $\dfrac{2\pi}{\pi/10} = 20$.

Similarly $\cos(A-B) - \cos(A+B) = 2\sin A\sin B$. With $A = \dfrac{\pi x}{30}$, $B = \dfrac{\pi}{3}$ the second factor is $2 \cdot \dfrac{\sqrt{3}}{2}\sin\dfrac{\pi x}{30} = \sqrt{3}\sin\dfrac{\pi x}{30}$, of period 60.

So $f(x) = 3\sin\dfrac{\pi x}{10}\sin\dfrac{\pi x}{30}$. Using $2\sin P\sin Q = \cos(P-Q) - \cos(P+Q)$: $f(x) = \dfrac{3}{2}\left(\cos\dfrac{\pi x}{15} - \cos\dfrac{2\pi x}{15}\right)$.

$\cos\dfrac{\pi x}{15}$ has period 30 and $\cos\dfrac{2\pi x}{15}$ has period 15. Two cosines with different frequencies cannot cancel, so T is a period of f exactly when it is a period of both, i.e. a multiple of 30.

Hence the fundamental period of f is 30. Directly: $\sin\left(\dfrac{\pi x}{10} + 3\pi\right) = -\sin\dfrac{\pi x}{10}$ and $\sin\left(\dfrac{\pi x}{30} + \pi\right) = -\sin\dfrac{\pi x}{30}$, so $f(x+30) = f(x)$.

The LCM of the periods of the two factors, 60, is a period of f but not the fundamental one, because both factors change sign after 30.

Final Answer: 30

Question 36

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \dfrac{15x\left(x^2+1\right)}{4x^4 + 17x^2 + 4}$. Let M and m be the greatest value and the least value of f, respectively. Then the value of $M - m$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{15x\left(x^2+1\right)}{4x^4 + 17x^2 + 4}$ द्वारा परिभाषित है। माना M तथा m क्रमशः f के अधिकतम मान तथा न्यूनतम मान हैं। तब $M - m$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f(-x) = -f(x)$, so f is odd, and $f(0) = 0$. It is enough to find the values of f for $x > 0$.

For $x > 0$, divide the numerator and the denominator by x^2: the numerator becomes $15\left(x + \dfrac{1}{x}\right)$ and the denominator $4\left(x^2 + \dfrac{1}{x^2}\right) + 17$.

Put $u = x + \dfrac{1}{x}$. Then $x^2 + \dfrac{1}{x^2} = u^2 - 2$, so the denominator is $4u^2 + 9$ and $f(x) = \dfrac{15u}{4u^2 + 9}$. By AM-GM, $u \geq 2$ for $x > 0$, with $u = 2$ only at $x = 1$, and u takes every value in $[2, \infty)$.

Write $f = \dfrac{15}{4u + \frac{9}{u}}$. The function $\psi(u) = 4u + \dfrac{9}{u}$ has $\psi'(u) = 4 - \dfrac{9}{u^2} > 0$ for $u > \dfrac{3}{2}$, so it is increasing on $[2, \infty)$, with least value $\psi(2) = \dfrac{25}{2}$ and $\psi(u) \to \infty$ as $u \to \infty$.

Hence for $x > 0$, $0 < f(x) \leq \dfrac{2 \cdot 15}{25} = \dfrac{6}{5}$, and $f(1) = \dfrac{30}{25} = \dfrac{6}{5}$. (Applying AM-GM directly, $\psi(u) \geq 12$, would give $\dfrac{5}{4}$, but equality needs $u = \dfrac{3}{2} < 2$, so $\dfrac{5}{4}$ is not a value of f.)

Since f is odd, $-\dfrac{6}{5} \leq f(x) < 0$ for $x < 0$ with $f(-1) = -\dfrac{6}{5}$. So f is bounded, its range is $\left[-\dfrac{6}{5}, \dfrac{6}{5}\right]$, $M = \dfrac{6}{5}$ and $m = -\dfrac{6}{5}$.

$M - m = \dfrac{12}{5} = 2.4$.

Final Answer: 2.4

Question 37

English

Let a be a positive real number and let $f : [1, 2] \to \mathbb{R}$ be defined by $f(x) = x + \dfrac{a}{x}$. If the least value of f is $\dfrac{9}{5}$, then the value of a is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना a एक धनात्मक वास्तविक संख्या है तथा $f : [1, 2] \to \mathbb{R}$, $f(x) = x + \dfrac{a}{x}$ द्वारा परिभाषित है। यदि f का न्यूनतम मान $\dfrac{9}{5}$ है, तो a का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

f is continuous on the closed, bounded interval $[1, 2]$, so it is bounded there and attains a greatest and a least value. The least value occurs at an end point $x = 1$, $x = 2$ or at an interior critical point.

$f'(x) = 1 - \dfrac{a}{x^2}$ vanishes only at $x = \sqrt{a}$ (as $x > 0$), and f' changes from negative to positive there, so $x = \sqrt{a}$ is a minimum point of f on $(0, \infty)$ with $f(\sqrt{a}) = 2\sqrt{a}$.

Case $1 \leq a \leq 4$: $\sqrt{a} \in [1, 2]$, so the least value of f on $[1, 2]$ is $2\sqrt{a} \geq 2$. It cannot equal $\dfrac{9}{5} < 2$. (Setting $2\sqrt{a} = \dfrac{9}{5}$ gives $a = \dfrac{81}{100}$, but then $\sqrt{a} = \dfrac{9}{10}$ lies outside $[1, 2]$, so this value is wrong.)

Case $a > 4$: $\sqrt{a} > 2$, $f'(x) < 0$ on $[1, 2]$, f is decreasing, and the least value is $f(2) = 2 + \frac{a}{2} > 4$. Impossible.

Case $0 < a < 1$: $\sqrt{a} < 1$, $f'(x) > 0$ on $[1, 2]$, f is increasing, and the least value is $f(1) = 1 + a$. Then $1 + a = \frac{9}{5}$ gives $a = \frac{4}{5}$, which indeed lies in $(0, 1)$.

Check: for $a = \frac{4}{5}$, $f(1) = \frac{9}{5}$, $f(2) = \frac{12}{5}$ and f is increasing on $[1, 2]$, so the least value is $\frac{9}{5}$.

Final Answer: 0.8